DE–3415 **11**

DISTANCE EDUCATION

P.G.D.C.A (S) DIPLOMA EXAMINATION, MAY 2012.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. (a) Write short notes on complements.

   (b) Convert the following :

   (i) $(FAFA.63)_{16} \rightarrow (\ )_8, (\ )_2$

   (ii) $(777.11)_8 \rightarrow (\ )_{10}$.

2. (a) State and prove De morgan's theorem.

   (b) Simplify : $xy + x\overline{y} + yz + xyz + x\overline{z}$.

3. Design a 8×1 multiplexer circuit.

4. Design the Parallel Adder circuit.

5. Discuss on

   (a) Clocked RS FF

   (b) Master – Slave FF .

6. Write a detailed notes on memory devices.

7. Draw the Flow chart for Instruction Cycle and explain the concept.

8. Expound on :

   (a) I/O Interface and

   (b) Programmed I/O.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. (a) Discuss on Binary codes.

   (b) Write a detailed notes on conversion from one base to another base.

10. (a) Explain about the theorems in Boolean Algebra.

    (b) Discuss about SOP and POS.

11. Simplify using K - map and Draw the logic circuit

    (a) $F(A, B, C, D) = \sum (0, 1, 2, 3, 4, 11, 12, 15)\ d\ (5, 6, 7, 14)$

    (b) $F(W, X, Y, Z) = \sum (2, 4, 6, 8, 10, 11, 13, 15)$

12. (a) Design the Full Adder circuit.

    (b) Discuss about Shift Registers.

13. (a) Expound on BCD counters.

    (b) Explain about Peripheral Devices.

14. Elucidate on the working of DMA.

15. Design ALU circuit.

DE–3416 **12**

DISTANCE EDUCATION

P.G.D.C.A.(S) DEGREE EXAMINATION, MAY 2012.

First Semester

C AND DATA STRUCTURES

(2011 Calender year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Discuss about constants and variables in C with examples.

2. Write about Nested Control Structures with examples.

3. What are Functions? How will you passing arguments to a function? Give examples.

4. What are Macros? Discuss them with examples.

5. What are pointers? Discuss about passing pointers to functions with examples.

6. Distinguish between Stack and Queue Data Structures.

7. List the advantages of Linked Lists.

8. Write a short note on: Doubly Linked List.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the various Branching and Looping statements in C with suitable examples.

10. Write a C program to find the sum and average of N given numbers.

11. Explain about C processor and Directives with examples.

12. Write a C program to create and process a data file.

13. Explain any ONE application of Stack with an example C program.

14. Discuss about Singly Linked Lists and their operations.

15. Explain about Trees and their applications with examples.

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DE–3417 **13**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, MAY 2012.

First Semester

RELATIONAL DATA BASE MANAGEMENT SYSTEMS

(2011 calender year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Mention the advantages of database management system.

2. How database differs from data file?

3. Discuss the concepts of relational model.

4. Explain the mapping operation in relational model.

5. What are the benefits of oracle?

6. List the characteristics of oracle users.

7. How to join multiple table through a query?

8. Explain the concept of using Indexes in tables.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the concept of Schema and subschema.

10. Discuss the manipulative capabilities available in database management system.

11. Discuss the following in relational model

    (a) Data manipulation facilities

    (b) Data definition facilities.

12. Describe the architecture for the oracle database.

13. Explain the properties of relational database and the components of RDMS.

14. With examples, show how queries are formed in oracle.

15. Discuss on embedded SQL statements.

DE–3418 **14**

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, MAY 2012.

First Semester

OPERATING SYSTEMS

(2011 calender year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions in about 200 words each.

1. Give the purpose of an Operating System.

2. Explain process state transitions.

3. What are the advantages and disadvantages of fixed partitioning?

4. Explain FIFO scheduling.

5. List the advantages of Virtual Memory.

6. Explain the semaphore operation.

7. Describe Disk scheduling process.

8. Explain page replacement algorithms.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. How GUI based operating system differs from CUI based operating system?

10. Explain scheduling algorithms in detail.

11. Explain the following:

    (a) Process scheduling

    (b) Priority scheduling.

12. What is deadlock? How to detect and prevent it.

13. Briefly explain file discriptor.

14. Briefly explain fault tolerance techniques.

15. Discuss protection mechanism available in file management.

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DE–3419 **21**

DISTANCE EDUCATION

P.G.D.C.A (S) DIPLOMA EXAMINATION, MAY 2012.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. What is the fundamental idea behind object oriented programming languages? Describe the various applications of object–oriented programming.

2. What are tokens in C++? Explain briefly.

3. Write a C++ program to find the factorial of a given number.

4. Discuss inline function with an example.

5. Write about general form of a class declaration and how to create objects for a class.

6. Define : constructor. Write down the characteristics of constructor functions.

7. Write a C++ program to implement the concept of operator overloading.

8. Explain the various file mode options available in C++.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. What are the types of operators supported by C++? Explain briefly with examples.

10. Write a C++ program using structures to create student's grade sheet.

11. Explain the need for friend functions.

12. Describe any three control structures in C++ with examples.

13. What are the different types of inheritance? Explain with examples.

14. Explain virtual functions in detail.

15. Write a C++ program to read the bio–data from file using command line arguments.

DE–3420 **22**

DISTANCE EDUCATION

P.G.D.C.A. (S) DEGREE EXAMINATION, MAY 2012.

Second Semester

UNIX AND SHELL PROGRAMMING

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — ($5 \times 8 = 40$ marks)

Answer any FIVE questions.

1. Explain the directory hierarchy.

2. What are kinds of permissions for the file? How to change it? Explain.

3. What is Meta characters? Explain.

4. Explain the trap command.

5. How to get and put tracking file changes?

6. Explain any four string functions.

7. What is meant by Error Recovery?

8. Define Built in functions.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

9. Explain the group command with various options.

10. Explain the sort, commands with various options.

11. Write a Unix program to sorting string in alphabetical order.

12. What is meant by shell variables? Explain.

13. (a) Explain the stream editor.

    (b) Write a shell program to check a given string is palindrom (or) not.

14. Describe the UNIX environment.

15. Explain the functions and procedures with an example.

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DE–3421 **23**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

Second Semester

COMMUNICATION SKILLS

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss in detail about objectives and barriers of effective communication.

2. Explain about Agenda writing and also discuss about when to write and when not to write.

3. Write short notes on :

   (a) Telephone conversation

   (b) Advantages of listening.

4. Explain how the GD is used to test the personality of a person.

5. Explain how the body language decides the attitude of a person.

6. Write short notes on the meaning and importance of meetings.

7. What is Negotiation and give the importance of Negotiation?

8. What is presentation and the use of notes and outlines?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail about the reading skill development techniques.

10. Explain about the modes and types of communication.

11. Explain how will you do the following :

    (a) Present a report

    (b) Asking questions

    (c) Express yourself

    (d) Writing articles.

12. (a) What is the importance of GD? How it acts as an important tool in conducting interviews?

    (b) Elucidate on the communication skills for meetings and employee communication.

13. Discuss on :

    (a) Mock meetings

    (b) Mock Interview

    (c) Mock Negotiation.

14. Write short notes on :

    (a) Communication in management

    (b) Within management

    (c) By specialists.

15. Elaborately discuss about presentation.

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DE–3422 **24**

DISTANCE EDUCATION

P.G.D.C.A (S) EXAMINATION, MAY 2012.

Second Semester

COMPUTER GRAPHICS

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions in about 200 words each.

1. With neat diagrams, explain the workings of input devices, "Light pen" and "Mouse".

2. Discuss on any region filling techniques.

3. Derive the rotation (Clockwise and anti-clockwise) transformation with neat diagram.

4. Give a note on two-dimensional viewing transformations.

5. With matrix representation, discuss the three dimensional scaling transformation.

6. With neat diagram, explain the three dimensional viewing process.

7. Give a note on hidden surface algorithm.

8. Write about the importance of command language with suitable illustrations.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions in about 400 words each.

9. Compare and contrast DDA line drawing method with Bresenham's line drawing procedure.

10. Explain the procedure of midpoint circle generation technique with algorithm.

11. Describe the cohen-sutherland line clipping algorithm with procedure.

12. Narrate the concept of Sutherland_hodgeman polygon clipping method with program segment.

13. Explain the rotation transformation of a three dimensional object with matrix values.

14. Elucidate the Z-buffer algorithm and explain in detail.

15. List and explain the various styles of command language.

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DE–4342 **15**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

First Semester

LAB I — DATA STRUCTURES USING C

(2011 Calendar Year onwards)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C program to find the roots of the quadratic equation $x^2 - 5x + 6 = 0$.

   (b) Write a C program to implement push and pop operations on stack.

---------- Cut here ----------

2. (a) Write a C program to multiply two matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

   (b) Write a C program to evaluate the mathematical expression $4 * 3 / 2 + 5 - 15 / 3$ using stack.

---------- Cut here ----------

3. (a) Write a C program to find the factorial value of 5 using recursion.

   (b) Write a C program to implement insert and delete operations on queue using array concept.

4. (a) Write a C program to check the word MALAYALAM is Palindrome or not.

(b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

5. (a) Write a C program to copy the contents of one file to another file.

(b) Write a C program to sort the following numbers in ascending order sequence :

90, 80, 70, 60, 50, 40, 30, 20, 10.

DISTANCE EDUCATION

P.G. D.C.A. (S) EXAMINATION, MAY 2012.

First Semester

LAB II — RDBMS

(Relational Data Base Management Systems)

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by LOT system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

---------- Cut here ----------

1. A daily sales file contains records with the following fields :

   Department number, data, item description sale price of each item, quality, cost of each item

   Write a program using oracle to list all the impact data and compute total amount of sales and profit for each sales. Accumulate and print total of sales and profit. Show the output report in the following format.

Department number

Item description

Sales price

Quantity

Cost

Profit.

---------- Cut here ----------

2. A hospital maintains blood donors record information in a file. The file contains the following fields.

   Donor number, Donor name, Donor age, Donor address, Pin code, Place of birth, Blood groups (A, B, AB, & O)

   Write a program using oracle to print out the number name and address of the Donors for the following categories.

   (a) Blood donors having AB blood group

   (b) Blood donors in the age group between 16 to 25.

   (c) Female donors with 'A' blood groups in the age between 18 to 25.

---------- Cut here ----------

3. A news paper vendor buys newspapers on wholesale from a distributor for Rs. 4 and sell them in retail for Rs. 4.50. At the end of the day the unsold papers are returned to the distributor for a Rs. 1 rebate per paper. Write a program using oracle to prepare a report for the newspaper render in the following format with 9 week data

Week	Bought	Sold	Return	Profit	Loss
1	100	50	xx	xxx	xxx
2	75	60	xx	xxx	xxx
– –	– –	– –	– –	– –	– –
– –	– –	– –	– –	– –	– –
Total					

---------- Cut here ----------

4. Write a program in SQL from hospital billing system with the following fields.

   Patient No. Patient Name, Age, Doctor, Patient-Type (IN/OUT), consulting charge, Blood test charge, X-Ray charge, other test charge and total fee.

   Prepare reports with the following condition :

   (a) The patient undergone blood test.

   (b) The patient who have taken X-rays.

   (c) The patient, who belong in In- patient category.

---------- Cut here ----------

5. An examination has been conducted to a class of 55 students and the following are givens as inputs.

   Three test marks, register number and candidate name

   Write an SQL program to do the following.

   Assign a letter grade to each student based on the average score and list register number, average score grader. The minimum pass for each subject is 50.

Average score	Grade
90-100	A
75-89	B
60-74	C
50-59	D
0-49	Fail

DE-4344 **25**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

Second Semester

LAB III — GRAPHICS USING C++

(2011 Calendar year onwards)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by LOT system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flow chart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

---------- Cut here ----------

1. Develop a C++ program to manipulate the following transformations :

   (a) Draw a triangle ABC translate it to a distance by 5 units to the right and rotate in clockwise by about 45°.

   (b) Scale the object by 5 times.

---------- Cut here ----------

2. Write a C++ program to draw non-linear objects arcs and curves.

3. Write a C++ program to draw non- linear object circle and three concentric circle.

---------- Cut here ----------

4. Develop a C++ program to implement the Bresenham's algorithm for drawing circle and ellipse using graphics mode.

---------- Cut here ----------

5. Develop a C++ program to draw a straight line using Bresenham's line drawing algorithm.

---------- Cut here ----------

6. Write a C++ program to show graphical transformations of a two dimensional object.

---------- Cut here ----------

7. Develop a C++ program with graphics mode to implement polygon clipping algorithm.

---------- Cut here ----------

8. Create a base class shape, which contains the common elements of circle, triangle and rectangle, from which you derive three classes called ball, rect and tria.

   Include member functions set ( ) to set coordinates and draw ( ) which draws them. Write a main ( ) program to exercise this class using graphic mode and graphic functions.

---------- Cut here ----------

9. Write a C++ program to draw a straight line using simple DDA algoithm.

---------- Cut here ----------

10. Develop a C++ program to manipulate the graphical object using “SHADING” technique.

DE–4345 **26**

DISTANCE EDUCATION

P.G.D.C.A. (S) EXAMINATION, MAY 2012.

Second Semester

LAB IV – UNIX AND SHELL PROGRAMMING

(2011 calendar year onwards)

Time : Three hours Maximum : 100 marks

Break-up of marks

Record : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

ONE question should be selected by the Examiner and given to the candidates by lot system.

1. (a) A hospital maintains patient details; Write a shell program to list the following

   (i) Patients in a blood groups wise

   (ii) Patients in age between 20 to 30

   (iii) Exit.

   (b) Write a program for our-function calculator by using yacc command.

2. (a) Write a shell program to prepare electric bill for domestic consumers.

For first 100 units – Rs. 0.75/unit

For next 100 units – Rs. 1.50/unit

Above 200 units – Rs. 3.00/unit.

Prepare the Electricity bill in the following format

Customer no

Customer name

Previous reading

Current reading

Units consumed

Charge

(b) Write and execute a shell program to generate leave status report of an organization.

---------- Cut here ----------

3. (a) Write and execute a shell program to do the following

(i) Convert given string in center of given row in BOLD, normal or reverse video

(ii) Write given string at given row, column an BOLD, normal and reverse video

(iii) Search a given string using awk with various options.

(b) Write and execute a shell program to print monthly pay slips for employees in an organization. Assume your own data and generate accordingly.

4. (a) Write a Menu Driven shell program to copy, edit, rename, delete a file.

(b) Write a shell program to prepare electric bill for domestic consumers

For first 100 units – Rs. 0.75/unit

For next 100 units – Rs. 1.50/unit

Above 200 units – Rs. 3.00/unit

Prepare the Electricity bill in the following format

Customer no

Customer name

Previous reading

Current reading

Units consumed

Charge.

---------- Cut here ----------

5. (a) Write given string at given row, column an BOLD, normal and reverse video.

(b) A hospital maintains patient details; Write a shell program to list the following

(i) Patients in a blood groups wise

(ii) Patients in age between 20 to 30

(iii) Exit.

6. (a) Write and execute a shell program to send mail to groups of user y extracting their ids from etc/group file.

(b) Write a shell program which accepts the name of a file from the standard input and then perform the following tests on it.

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List unsorted and sorted file

(iv) Quit.